

Prerna

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (CGPA - 9.19/10.00)

Bhopal, India

Sep 2023 – May 2027

TECHNICAL SKILLS

Languages: Python, MySQL

Data Analytics: Excel, Power BI, Tableau, Matplotlib, Seaborn, Pandas, NumPy, statistical Analysis, EDA, ETL, KPI Analysis, Data Interpretation

Machine Learning: Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation

Tools: Git, GitHub, VS Code

Core Subjects: Operating System, Computer Networks, DataBase and Management Systems

EXPERIENCE

Information System Intern

Indian Oil Corporation Limited

May 2026 – June 2026

Barauni Refinery, India

- Developed technology solutions for the IS Department to streamline internal workflows and improve efficiency.
- Analyzed operational requirements and converted them into technical solutions for internal workflow automation.
- Prepared technical documentation and process reports supporting future maintenance and project scalability.

Machine Learning Intern

Zidio Development

May 2025 – Jun 2025

Remote

- Worked on data preprocessing and exploratory data analysis (EDA) using Python and Pandas to clean, organize and understand datasets.
- Evaluated basic Machine Learning models using algorithms such as Linear Regression, Logistic Regression, Decision Tree and Random Forest.
- Compared model performance using evaluation metrics such as accuracy, precision and recall.

Data Science Intern

Coincent AI

Apr 2024 – Jun 2024

IIT Madras

- Collected, cleaned and preprocessed structured datasets using Python, Pandas, and NumPy to improve data quality for analysis and model development.
- Performed Exploratory Data Analysis (EDA) using Python, Pandas, and Matplotlib to identify trends, correlations, and actionable business insights through statistical visualization.
- Collaborated with mentors and team members to translate business problems into machine learning solutions while following industry best practices.

PROJECTS

FlowCare Analytics - Hospital Patient Flow & Operational Intelligence Platform

Python, Pandas, MYSQL, Power BI, Scikit-learn

June 2026

- Built an end-to-end hospital operations analytics platform using 50K+ synthetic patient journeys to analyze waiting times, bottlenecks, resource utilization, and capacity loss.
- Designed a MySQL relational database to identify department-level bottlenecks and operational trends.
- Developed Power BI dashboards and a custom Patient Flow Bottleneck Score, with statistical analysis and ML-based congestion prediction to support operational decision-making.

Smart RAG Assistant with Zylos AI | LangChain, Sentence Transformers, FAISS

May 2026

- Developed a Retrieval-Augmented Generation (RAG) chatbot using LangChain, FAISS, Hugging Face Embeddings and Google Gemini, enabling intelligent answering from PDF documents and website content.
- Processed and indexed unstructured PDF and website data using LangChain and FAISS to enable intelligent information retrieval.

CERTIFICATIONS

Applied Machine Learning in Python - (Link)

Data Science with Python - (Link)

Machine Learning A-Z: AI, Python + ChatGPT Prize - (Link)

Scaler certified C++ - (Link)

Ethnus Certified Mern Full Stack - (Link)